

Zhiling Zhou

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Education

Yale University

Aug 2025 – Present

PhD Student, Department of Statistics and Data Science

Fudan University

Sept 2020 – Jun 2025

B.S. in Data Science and Big Data Technology

School of Data Science, Shanghai, China

Publications and Preprints

1. **How Transformers Represent Hierarchies: A Local-to-Global Mechanism.**

Zhiling Zhou, Tianhao Wang, Zhuoran Yang.

ICML 2026. [Paper Link](#)

2. **Demystifying Classifier-Free Guidance for Auto-Regressive Image Generation.**

Zhiling Zhou^{*}, Jiachun Pan^{*}, Fengzhuo Zhang^{*}, Dirk Bergemann, Zhuoran Yang.

Mechanistic Interpretability Workshop at ICML 2026. [Paper Link](#)

3. **Incremental Gauss–Newton Methods with Superlinear Convergence Rates.**

Zhiling Zhou, Zhuanghua Liu, Chengchang Liu, Luo Luo.

Journal of Optimization Theory and Applications. [Paper Link](#)

4. **DiffSim: Aligning Diffusion Model and Molecular Dynamics Simulation for Blind Docking.**

Yanlin Fei, Yurui Li, Zhiling Zhou, Haohan Wang.

IEEE BIBM 2024. [Paper Link](#)

5. **Robust Network Learning via Inverse Scale Variational Sparsification.**

Zhiling Zhou^{*}, Zirui Liu^{*}, Chengming Xu, Yanwei Fu, Xinwei Sun.

Preprint. [Paper Link](#)

Research Experience

How Transformers Represent Hierarchical Language Structure

Yale University

Advisors: Tianhao Wang, Zhuoran Yang

- Bridge Transformer-based context-free grammar generation with classical parsing algorithms.
- Develop a theoretical framework showing how Transformers implement parsing algorithm.

Demystifying Classifier-Free Guidance for Auto-Regressive Image Generation

Yale University

Advisors: Zhuoran Yang

- Demonstrated the repetition phenomenon and visual degradation in AR models lacking CFG.
- Theoretically modeled how image inductive bias shapes training dynamics and feature representation.

- Submitted to NeurIPS 2026. Preprint version to appear on arXiv.

Incremental Gauss–Newton Method for Nonlinear Equations

Fudan University

Advisor: Luo Luo

- Proposed an incremental Gauss–Newton method with superlinear convergence rates.
- Extended the method with mini-batch updates to improve convergence speed.

Robust Network Learning via Image Sparsification

Fudan University

Advisors: Xinwei Sun, Yanwei Fu

- Developed algorithms based on total variation and efficient Lasso optimization.
- Proposed a graph-based numerical acceleration technique.
- Improved robustness and interpretability of visual learning systems.

Aligning Diffusion Models and Simulations for Blind Docking

University of Illinois Urbana-Champaign

Advisor: Haohan Wang

- Aligned diffusion models with molecular dynamics simulations for blind docking.
- Constructed reliable blind docking datasets using molecular simulations.

Awards

- Third Prize of Excellent Student Scholarship, Fudan University (2020, 2022)
- KLA Scholarship, Fudan University & KLA Corporation (Top 4 undergraduates per year), 2024
- Outstanding Undergraduate Thesis Award, Fudan University, 2025.